Cell Systems

Supplemental Information

Pseudo-transition Analysis Identifies the Key Regulators of Dynamic Metabolic Adaptations from Steady-State Data

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Supplemental Figures

Acetate
Gluconate
Acetate
Glycerol
Acetate
Glucose
Acetate
Fructose
Acetate
Galactose
Acetate
Succinate
Acetate
Pyruvate
Galactose
Succinate
Gluconate
Pyruvate
Glycerol
Pyruvate
Glucose
Pyruvate
Fructose
Pyruvate
Galactose
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Gluconate
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Succinate
Glycerol
Gluconate
Glucose
Gluconate
Glucose
Glycerol
Fructose
Gluconate
Fructose
Glycerol
Fructose
Glucose
Galactose
Gluconate
Flux log changes
0.5
1
2
4
ρ
<-1
0
1
>2
n/a

Figure S1. 28 maps for regulation of flux changes of all pseudo-transitions, related to Fig. 3. The figure is composed as in Fig. 3B. For each of the 28 pseudo-transitions, regulatory coefficients for transcriptional and reactant regulation (ρ_{er}) are shown as a breakdown of the individual transcriptional (ρ_e), thermodynamic ($\rho_{\Delta G}$) and substrate kinetic (ρ_s) regulation on the metabolic network as in Fig. 2A. Log fold changes in flux between the two compared steady state conditions are shown by the thickness of the corresponding metabolic reaction.

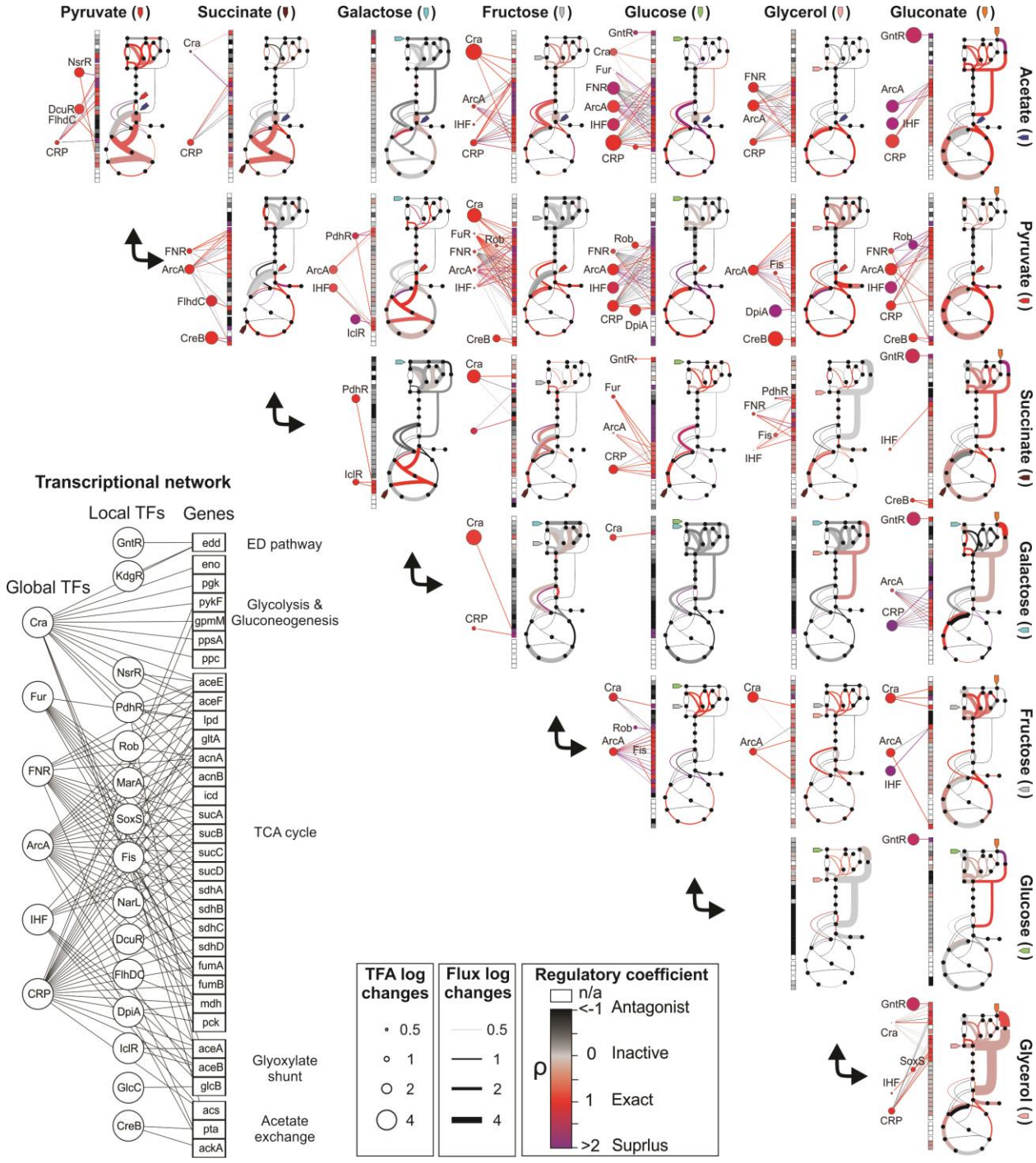


Figure S2. 28 maps for transcriptional regulation of flux changes of all pseudo-transitions, related to Fig. 6. The figure is composed as in Fig. 6 of the main text and shows all the 28 pseudo-transitions.

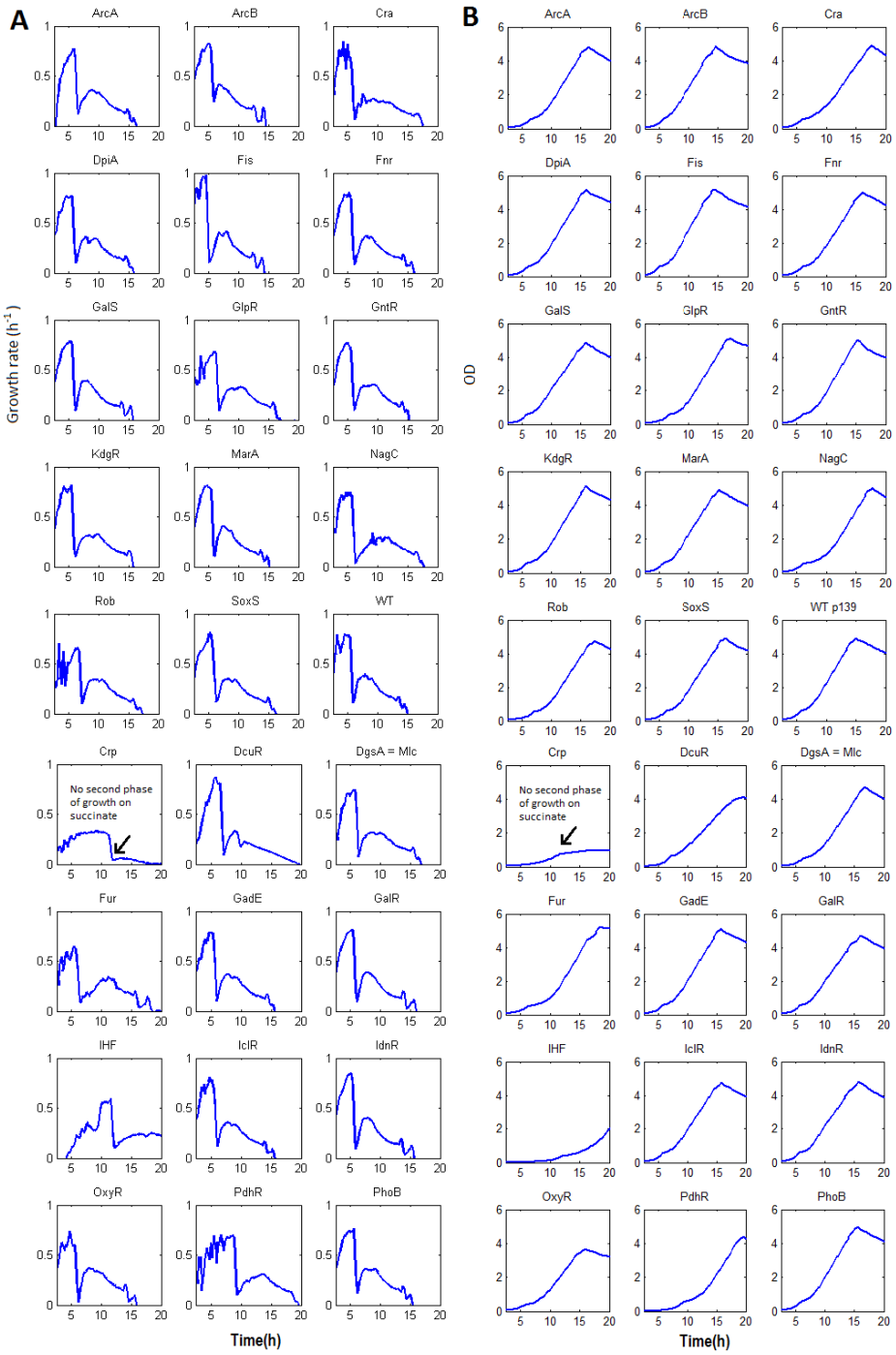


Figure S3. Dynamic growth of single transcription factor knockouts in the glucose and succinate medium, related to Fig. 7. A) Time-course growth rate of 26 single transcription factor knockout strains in the glucose and succinate medium. The transcription factor's name is shown on top of the corresponding growth curve (WT: wild type). CRP is the only transcription factor deletion abolishing the second growth phase on succinate. B) The same showing the Optical Density.

Supplemental Datasets

Dataset S1. Measurements and estimates for the steady state data in the eight different carbon sources. Contains: physiology, metabolic fluxes, metabolite concentrations, thermodynamic potentials, transcriptional network activity, kinetic orders and regulation coefficients.

Dataset S2. Measurements and estimates for the diauxic shift from glucose to succinate. Contains: physiology, metabolic fluxes, metabolite concentrations, thermodynamic potentials, GFP promoter reporters, regulation coefficients.

Dataset S3. Stoichiometric model of *E. coli* central metabolism.

Dataset S4. Log-log plots for all reactions and regulatory mechanisms for the eight steady states.

Dataset S5. Log-log plots for all reactions and regulatory mechanisms during the diauxic shift.

Extended Experimental Procedures

Strains, growth conditions and cultivation

All experiments were performed with *E. coli* BW25113 wild-type (Baba, Ara et al. 2006). Steady state physiology, ^{13}C-labelling, metabolite and RNA levels were obtained from exponentially growing cultures (**Dataset S1**). Cultivation was done in 35 ml M9 minimal medium with 5 g/L of one of the eight carbon source in 500 ml shake flasks at 37°C, 300 rpm, and a shaking diameter of 5 cm. Frozen glycerol stocks were used to inoculate Luria-Bertani (LB) complex medium. After 6 hours of incubation at 37°C and constant shaking, LB cultures were used to inoculate M9 medium precultures with the indicated carbon sources for overnight cultivation. Final cultures were inoculated 1:100 (v/v) with the same carbon source the next day.

For the diauxic shift experiment, the same inoculation scheme was used and physiology and metabolite levels were determined in 50 ml M9 medium cultures with 0.5 g/l glucose and 5 g/l succinate in 500 ml shake flasks at 37°C, 300 rpm, and a shaking diameter of 5 cm. Single TF knockout strains and GFP-promoter reporter strains were cultured in 96 deep-well plates (Kuehner AG, Birsfeld, Switzerland). Deep-well plates with minimal medium and glucose as the sole carbon source were inoculated 1:50 from LB precultures and incubated overnight at 37°C under shaking. Subsequently, 96 well flat transparent plates (Nunc, Roskilde, Denmark) containing M9 medium (fill volume 200 µL) with 0.5 g/l glucose and 5 g/l succinate were inoculated 1:200 with overnight cultures and sealed with parafilm to reduce evaporation. GFP-

promoter strains were cultivated and monitored on-line during the shift at 37°C with shaking using a plate reader (TECAN infinite M200, Tecan Group Ltd., Männedorf, Switzerland).

Determination of growth physiology

Cell growth in shake flask was monitored by determining the optical density at 600 nm (OD_{600}) using a spectrophotometer (Spectra Max Plus, Molecular Devices, Sunnyvale, CA). For steady state experiments, growth rates were determined by log-linear regression of OD_{600} during the exponential phase from at least four data points. Extracellular carbon accumulation and depletion were determined using HPLC (Heer and Sauer, 2008). Carbon uptake and secretion rates were determined from at least three biological replicates of independent shake flask experiments from at least four points in the exponential phase. Cell dry weight (CDW) was calculated from liter per OD_{600} using pre-determined conversion factors for each carbon source (**Dataset S1**). The biomass yield was calculated as the inverse of the slope of concentration against cell dry weight. The non-inversed slopes were further multiplied with the growth rate to obtain uptake and secretion rates.

For the diauxic shift, time-dependent growth rates were calculated by two-point finite difference numerical approximation of the natural logarithm of OD_{600} from two independent shake flask experiments. Cell dry weight was calculated from OD_{600} using the conversion factor estimated for growth on glucose (**Dataset S1**). Analytical functions were fit to time-courses of external metabolite concentrations to obtain carbon uptake and secretion rates by two-point finite difference numerical approximation divided by the corresponding cell dry weight (**Dataset S2**).

Metabolic flux analysis

For steady state analyses, separate ^{13}C-labeling experiments were performed with a mixture of 20% (wt/wt) [U-^{13}C] labeled isotopologue (>99%; Cambridge Isotope Laboratories, Andover, MA) and 80% (wt/wt) of natural abundance carbon sources. Separate ^{13}C-labeling experiments were performed with 100% [1-^{13}C]galactose, [1-^{13}C]glucose, [1-^{13}C]gluconate and [1-^{13}C]fructose (>99%; Cambridge Isotope Laboratories, Andover, MA) and [1,3-^{13}C]glycerol (>99%; CortecNet Voisins-Le-Bretonneux, France). Aliquots of fractionally ^{13}C-labelled biomass were prepared from exponentially growing cultures and analyzed by gas chromatography mass spectrometry (GC-MS)(Zamboni et al., 2009).

Estimation of absolute fluxes was done by whole isotopologue balancing (Kleijn et al., 2005; van Winden et al., 2005), using cumomer balances and cumomer to isotopologue mapping matrices (Wiechert et al., 1999) to calculate isotopologue partitioning of metabolites in a pre-defined stoichiometric network model for a given flux set. The flux set giving the best correspondence between measured and simulated ^{13}C-label partitioning and physiology measurements of

growth and extracellular fluxes (**Dataset S1**) was determined by non-linear optimization and selected as the final flux distribution. Standard deviations for metabolic fluxes were estimated through Monte Carlo simulations by re-estimating fluxes after adding Gaussian noise to the measured ^{13}C-labeling data (Schmidt et al., 1999) (**Dataset S1**).
Dynamic flux changes during the diauxic shift were estimated by flux balance analysis using the COBRA Toolbox (Schellenberger et al., 2011) and a stoichiometric model of central metabolism (**Dataset S3**). Fluxes were estimated constraining the model by the estimated carbon uptake rates, carbon secretion rates and growth rate (**Dataset S2**) under minimization of sum of fluxes and assuming pseudo-steady state for each time point. Lower and upper bound for exchange fluxes and biomass were set to one standard deviation of replicates. Upper and lower bounds for flux estimates were calculated by flux variability analysis.

Intracellular metabolite concentrations by LC-MS/MS

For steady state analysis, 1 ml aliquots were taken in a 37°C room from exponential phase cultures. After vacuum-filtration on a 0.45 µm pore size nitrocellulose filter (Millipore), samples were immediately washed with two volumes of fresh 37°C M9 medium containing the respective carbon source at a pH adjusted to the value of the culture broth at the time of sampling. Subsequently filters were directly transferred for extraction into 4 ml of 60% (v/v) ethanol/H_2O to which 100 µl of internal standard (fully ^{13}C-labelled *Saccharomyces cerevisiae* extract) was added and incubated at 78°C for 2 min. For the diauxic shift, 2 ml aliquots were withdrawn at 9 different points and vacuum-filtered on a 0.45 µm pore size nitrocellulose filter (Millipore). Filters were directly subjected to cold extraction (-20°C) with 40:40:20 acetonitrile/methanol/water containing 200 µl of internal standard (fully ^{13}C-labelled *S. cerevisiae* extract). In both cases, extracts were separated from the filters and residual cell debris and nitrocellulose was removed by centrifugation. Cell extracts were thawed, dried at 120 µbar, and resuspended in 100 µl deionized H_2O of which 15 µl were transferred into rubber-sealed HPLC tubes. Metabolite abundances were determined by ion-pairing ultra-high performance liquid chromatography (UPLC)-tandem MS (Buescher et al., 2010) and quantified through a dilution series of a mix containing all metabolites and internal standard. Intracellular metabolite concentrations in µmol/mL were calculated from metabolite abundances in µmol/gCDW using a previously determined conversion factor to intracellular cytosolic volume (**Dataset S1**).

Estimation of thermodynamic potentials

To estimate thermodynamic driving forces for each reaction in our network, we used a variant of the constraint-based method max-min driving force (Noor et al., 2014). Briefly, we set concentrations of measured metabolites allowing other metabolites to vary within physiological

ranges. We used the component contribution method (Noor et al., 2013) to calculate standard Gibbs energies and set the net flux direction for each reaction according to estimated metabolic fluxes. Finally, we applied the MDF optimization iteratively to maximize the driving forces in all reactions given the constraints and propagated measurement errors of metabolite concentrations to calculate upper and lower bound for each reaction. Infeasibilities, which accounted for only 4% (13/336) of the estimated ΔGs, were solved by iteratively relaxing the corresponding directionality constraints.

Current metabolomics methods are not sufficient for obtaining a complete picture of the changes in Gibbs free energies for all the reactions in central metabolism. The main hurdle is the incomplete quantification of several key metabolites (such as erythrose-4P, BPG, GAP, glyoxylate, and the individual concentrations of 2PG and 3PG). In order to obtain a full estimation of the driving force of all active reactions in the model, we assumed that these unknown concentrations can vary within a physiological range (10^{-6} - 10^{-2} M) and applied the optimization criterion which is based on the Max-min Driving Force principle (Noor et al., 2014). According to this principle, the concentrations of the unknown metabolites are adjusted to maximize the driving force of the least energetic reaction in the network. However, applying this principle only once typically leaves many of the metabolites' concentrations free as they do not limit the driving force of the bottleneck reaction. Here, we implemented an extension of MDF, called IMDF, that performs the optimization iteratively, each time fixing the concentrations of a subset of metabolites (and thus the driving force of the bottleneck reaction), and in the next step optimizing the driving forces of the remaining reactions which are not yet fixed. The algorithm terminates when the concentrations of all metabolites in the network become fixed - and these values are also the output of the method. Since all metabolites now have unique concentration, the driving forces of all the reactions in the model can be determined. We propagated the standard deviation measurement errors of metabolite concentrations appearing in each reaction to calculate the upper and lower bound for each reaction. The source code for IMDF can be obtained from GitHub (https://github.com/eladnoor/component-contribution)

The input paramters for IMDF were:

- Stoichiometric matrix: We use the same stoichiometric matrix as the one used for MFA in order to determine the absolute fluxes in the network.
- Standard Gibbs free energies: The component-contribution method (CCM) (Noor et al., 2013) is used to estimate the mean value and confidence interval of the Gibbs free energy in standard conditions - ΔG'°.
- Metabolite concentrations: Metabolomics measurements provide concentrations for some of the metabolites in the model (with confidence intervals estimated from the standard error of biological repeats). For those metabolites which are not measured by this method, the intervals are set to the general physiological range of 10^{-6} - 10^{-2} M.

- Flux directions: MDF does not require the actual values of the fluxes in the system, but the direction of flux is used for setting the constraints on the driving forces. Flux direction was set according to results from ^{13}C flux analysis.

Transcripts levels by microarrays

Aliquots of three independent cultures were harvested during exponential phase on each carbon source. RNA synthesis was blocked by adding 10% (v/v) stop solution (5% (v/v) TE-saturated-phenol in 95% ethanol) and samples were mixed and spun down for 10 min at 4°C. Supernatants were decanted and cell pellets redissolved in 200 µl of TE buffer (10 mM Tris·Cl, 1 mM EDTA, pH 8.0) containing 15 mg/ml lysozyme, vortexed for 10 s and incubated at room temperature (15–25°C) for 10 min under constant shaking. 700 µl of RLT buffer was added from the RNeasy Mini Kit (QIAGEN), samples were vortexed for 10 seconds then transferred to 2 ml Safe-Lock tubes containing acid-washed glass beads. Cells were mechanically disrupted by vortexing for 10 minutes at maximum speed. After centrifugation supernatants were transferred to a new tube and mixed with 470 µl ethanol. Total RNA was extracted from the bacterial lysates using the RNeasy Mini Kit (QIAGEN), followed by removing residual genomic DNA with the Turbo DNA-free Kit (Ambion). The microarray experiment used three independent RNA isolations from each strain that were pooled together with equal quantity. Gene expression was assessed by single color Agilent *E. coli* GE, 8x15k (020097) arrays (**Dataset S1**). After analysis by PCA, one replicate for growth on acetate was found to be substantially dissimilar from the other two and was removed from further analysis. Microarray data are available in the ArrayExpress database (www.ebi.ac.uk/arrayexpress) under accession number E-MTAB-3392.

Calculation of relative protein changes was based on the assumption that translation rates are not affected by the environmental perturbations. As described before (Chubukov et al., 2013), log fold changes in enzyme abundance (E) can thus be calculated from the measured transcript levels (T), growth rate-dependent total RNA (R) and measured growth rates:

$$\log\left(E_{ij}/E_{iz}\right) = \log\left(T_{ij}/T_{iz}\right) + \log\left(R(\mu_j)/R(\mu_z)\right) - \log\left(\mu_j/\mu_z\right)$$

where the index i refers to a gene and the index j and z to one of the eight conditions. The amount of mRNA per biomass was assumed to be a constant fraction of total RNA, while total RNA was assumed to be a function of the growth rate; an affine fit was calculated based on published data (Bremer and Dennis, 1996) (**Dataset S1**). For a large fraction of the transcripts the fold changes were within an order of magnitude of growth-dependent dilution, hence estimated protein abundances generally decreased with growth rate, in line with experimental evidence (Klumpp et al., 2009) (Gerosa et al., 2013).

Transcriptional network activity by network component analysis (NCA)

Transcription factor activities and control strengths were inferred through network component analysis (Liao et al., 2003) using the transcriptional network topology from RegulonDB (Salgado et al., 2013), updated with interactions from DNA-protein interaction screens (Shimada et al., 2010)(Shimada et al., 2011)(**Dataset S1**). To perform estimation on a not NCA-compliant topology, we employed a previously published stochastic implementation (Buescher et al., 2012) that was run 2500 times with random initial parameters. The best reconstruction, i.e. with the lowest sum of squared errors, was considered for analysis (**Dataset S1**). The reconstruction was fairly accurate with Pearson correlation between measured and reconstructed expression of 0.92 and with 77% of control strengths matching annotated regulatory modes of activation and repression.

Enzyme abundance during the shift by GFP-reporters

To measure enzyme abundance for the entire carbon metabolism, 37 strains of green fluorescent protein (GFP)-based promoter reporter plasmids were obtained from a library (Zaslaver et al., 2006) and an additional 23 were constructed by PCR following the procedures of the original study (**Dataset S2**). On-line measurements of OD_{600} and GFP fluorescence (excitation wavelength: 500 nm, emission wavelength: 530 nm) were performed using a plate reader (TECAN infinite M200, Tecan Group Ltd., Männedorf, Switzerland) at 10 min intervals and analyzed using custom MATLAB software to obtain growth rate (dln(OD)/dt) and expression profile (GFP/OD) as previously described (Gerosa et al., 2013). Hierarchical clustering of expression profiles was based on Pearson correlation and performed using the *pdist* and *linkage* MATLAB functions (**Dataset S2**).

Estimation of regulation coefficients

To obtain regulation coefficients, the stoichiometric model (**Dataset S3**) and the transcriptional topology (**Dataset S1**) were used to compile a list of flux-enzyme, flux-ΔG, flux-substrate and transcription factor-gene pairs as described in the main text. Regulation coefficients were calculated as the slope between fold changes of regulatory inputs and functional outputs between conditions as defined generally in **Eq. 1** and derived for regulation of metabolic fluxes in **Eq. 4** and for gene expression in **Eq. 6** Corresponding log-log plots are available in **Dataset S4**. For the diauxic shift, time-course metabolite concentrations and thermodynamic potentials were synchronized on the same sampling time of metabolic fluxes by linear interpolation. For each flux-enzyme, flux-ΔG and flux-substrate pair, overall regulation coefficients ($\overline{\rho}_{e_i}, \overline{\rho}_{\Delta G_i}, \overline{\rho}_{s_{ix}}$) were estimated by orthogonal regression over all the time-course data points. Kinetic orders

(α) used for calculation were the one inferred from steady state data (**Dataset S1**). Log-log plots for each flux-enzyme, flux-ΔG and flux-substrate pair during the shift are available in **Dataset S5**.

The standard deviations σ_ρ and $\sigma_{\sum \rho_x}$ for regulation coefficients of individual and aggregate regulatory inputs, respectively, comparing conditions j and z were calculated by error propagation of standard deviations for regulatory inputs (σ_I) and functional output (σ_O) measurements as derived from **Eq. 1**:

$$\sigma_{\alpha_x \cdot \Delta\log(\mathrm{I}_x)} = \alpha_x \cdot \sqrt{\left(\sigma_{I_{xj}}/I_{xj}\right)^2 + \left(\sigma_{I_{xz}}/I_{xz}\right)^2}$$

$$\sigma_{\Delta\log(\mathrm{O})} = \sqrt{\left(\sigma_{Oj}/O_j\right)^2 + \left(\sigma_{Oz}/O_z\right)^2} \qquad \textbf{Eq. MM1}$$

$$\sigma_{\rho_x} = \sqrt{\left(\frac{\sigma_{\alpha_x \cdot \Delta\log(\mathrm{I}_x)}}{\alpha_x \cdot \Delta\log(I_x)}\right)^2 + \left(\frac{\sigma_{\Delta\log(\mathrm{O})}}{\Delta\log(O)}\right)^2} \cdot \rho_x \quad \sigma_{\sum \rho_x} = \sqrt{\sum_x {\sigma_{\rho_x}}^2}$$

The standard error of measurement (SEM) was calculated from the standard deviations and the number of independent observations of measurements (n=3):

$$SEM = \sigma_\rho/\sqrt{n}$$

Estimation of kinetic orders

With enzyme abundance, thermodynamic potential and substrate concentrations available for multiple steady states, unknown kinetic orders (α) necessary to quantify substrate regulation (ρ_s) can be estimated by linear regression as an upper-bound in explaining the observed flux changes (Chubukov et al., 2013):

$$\min_{0 \le \alpha \le 5} \quad \log(J_i) - \log(E_i) - \log(\Delta G_i) = \sum_{x \in S_i} \alpha_{ix} \cdot \log(M_x) \qquad \textbf{Eq. MM2}$$

with α constrained to be between 0 and 5 to set a biologically realistic upper bound on the non-linear gain. We estimated kinetic orders independently for each flux-enzyme pair and flux directionality by least square optimization of **Eq. MM2** using the *lsqlin* function of MATLAB (**Dataset S1**).

Performances of differential change and functional consistency analyses quantified by ROC curves

For differential change analysis, reactions were classified as actively regulated when log fold changes in transcript levels exceeded a given cut-off C ($|\Delta\log_2(E)| \ge$C) for at least one of the enzymes. For functional consistency analysis, reactions were classified as actively regulated when the proportionality between log fold transcript and flux changes, as quantified by the regulation coefficient $\bar{\rho}$ estimated by orthogonal regression over all conditions ($\log(J) \cdot \bar{\rho} =$

$\log(E)$), was near unity within a certain displacement D $(1 - D \leq \bar{\rho} \leq 1 + \mathrm{D})$ for at least one of the enzymes. For each of the 28 pairwise comparisons, the cut-off C and the displacement D were varied to obtain classifications gradually ranging from all reactions being classified as not regulated to all reactions being regulated. The gold standard of *pseudo*-transition analysis was obtained by classifying each reaction as actively regulated if at least one of its enzymes explained flux changes within a factor of two $(0.5 \leq \rho_e \leq 2)$. True and false positive rates of classification by differential change and functional consistency analysis were thus calculated as the fraction of reactions correctly/incorrectly identified as actively regulated with respect to the gold standard. To ensure independence from the selected cut-offs, the same operations were repeated using three alternative cut-off sets for *pseudo*-transition analysis $(0.8 \leq \rho_e \leq 1.2\ , 0.3 \leq \rho_e \leq 3, 0.25 \leq \rho_e \leq 4)$. The obtained ROC curves were used to define the boundaries for true and false positive rates.

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